MACROECONOMICS

MACRO-12

Time: 5 minutes

Outcome: Analyze physical capital, human capital, natural resources, and technology.

Difficulty: Intermediate

Sources of Productivity

THE CORE IDEA

Productivity rises with physical capital, human capital, natural resources, and technological knowledge. Capital can have high returns where it is scarce. More capital per worker can raise output per worker, but the production function becomes flatter as capital deepens. Productivity compares output with the labor input used to produce it, so changes in both output and labor matter.

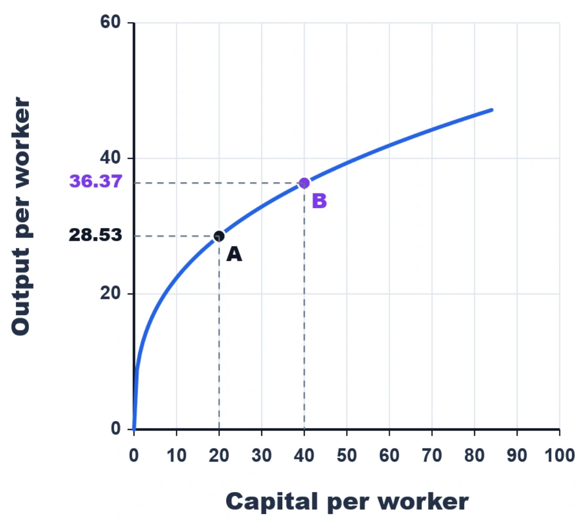
HOW TO RECOGNIZE IT

- A worker receives a new machine that raises output with the same labor time.
- More capital per worker can raise output per worker, but the production function becomes flatter as capital deepens.
- The graph is upward sloping but becomes flatter as capital per worker rises.
- A low-capital country buys machinery, but workers lack training and firms cannot reliably obtain electricity.

WATCH OUT

Do not treat every increase in capital as producing the same output gain. More capital per worker can raise output per worker, but the production function becomes flatter as capital deepens.

WORKED EXAMPLE: CAPITAL DEEPENING AND PRODUCTIVITY



Increasing capital per worker from 20 to 40 raises output per worker from 28.53 to 36.37, a gain of 7.84. The production function becomes flatter as capital deepens, so equal additions to capital generate progressively smaller output gains. That curvature shows diminishing returns to capital.

CHECK YOURSELF

How does a worker's machine differ from the worker's training as a source of productivity?

READY? Return to the game and master this concept.